Ayush Kumar Bhadani

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Education

Bangalore Institute of Technology **2023-2027**
Bachelor of Technology in Computer Science and Engineering (Data Science) | ***CGPA**: 9.45* *Bangalore, Karnataka*

Experience

AI Research Intern **Oct 2026 – Present**
Siemens Technology and Services Pvt. Ltd. *Bangalore, India*

- **Conducting applied research** in **AI-driven autonomous software engineering**, studying how systems can adapt and improve themselves within defined **safety and correctness** constraints.
- **Surveying** recent literature and **open-source LLM-driven systems** to compare approaches.
- **Analyzing** how specifications, decision records, and static rules define **correctness boundaries** for automated changes.
- **Designing evaluation and verification approaches**, including a small demo application, to test AI-driven system behavior against explicit correctness criteria.
- **Presenting** findings and technical comparisons to mentors and engineering leads through discussions and notes.

Projects

CodeVerdict | *Python, Generative AI, Agentic AI, RAG, FAISS, FastAPI, REST API, Docker, Groq/Llama 3* **Live App**

- Architected an **autonomous multi-agent GenAI system** for **AI-driven software quality assurance**, orchestrating **4 specialized agents** (Security, Quality, Performance, Test Coverage) as a stateful **graph-based** workflow with explicit state management and conditional execution, running concurrently on Python **asyncio**.
- Developed a **RAG knowledge-retrieval layer** using **Sentence Transformer** embeddings and **FAISS vector database** semantic search to ground **LLM** outputs in secure-coding docs; integrated **Bandit SAST** via a deterministic merge-verification step, exposed through **scalable REST APIs**, and **containerized the service with Docker**.
- Designed an **experimentation, automated testing and benchmarking framework** on a labeled PR dataset, achieving **87.5% precision** and **100% recall**, with documented failure-mode analysis driving iteration.

Digital Wellbeing Intelligence System | *Python, scikit-learn, FastAPI, Plotly, NLP, Pandas/NumPy* **Live App**

- Engineered a **data science and machine learning pipeline** (data prep, feature engineering, training, evaluation) using **K-Means clustering** to segment users into **3 behavioral risk profiles** at ∼85% consistency.
- Engineered **4 behavioral features** from structured and unstructured data, applying normalization to cut intra-cluster variance by ∼**30%** and stabilize segmentation.
- Deployed **async FastAPI REST web services** to a **cloud platform** at ∼**200ms** latency over **10K+ records**; replaced manual reporting with automated **Plotly telemetry dashboards**, cutting reporting effort ∼**40%**.

Technical Skills

Programming: Python, C, C++, SQL, OOP
Generative AI & LLMs: Multi-Agent Systems, RAG, Prompt Engineering, Sentence Transformers, LLM APIs (Groq, Llama 3)
AI-Assisted Development: GitHub Copilot, Claude Code, GenAI-assisted SDLC
AI & Machine Learning: Data Science, Deep Learning (CNNs, PyTorch), K-Means, NLP, scikit-learn, Model Evaluation
Software Engineering: Software Architecture, Design Patterns, asyncio, FastAPI, REST APIs, Scalable Systems, SDLC
Cloud & Platform Engineering: Docker, Cloud Deployment (Vercel, Render), Git, GitHub, Agile
Data & Databases: SQL, MySQL, MongoDB, FAISS (Vector Database), Pandas, NumPy, Plotly
Relevant Coursework: DSA, DBMS, OS, Computer Networks, AI/ML

Achievements

- **Research Publication**: Co-authored a research paper on **CNN-based binary image classification**; trained a **PyTorch** CNN on the ISIC dermoscopic dataset, reaching **98% accuracy**. *View Paper*
- **Academic Excellence**: Ranked highest in **CSE (Data Science)** at BIT, with a perfect **10.0 CGPA** in the 6th semester and **9.86** in the 3rd.

Leadership & Activities

Core Technical Member | *Google Developer Group (GDG)*

- Implemented the frontend UI of the official GDG **web application** with a **5-member team** in **Agile sprints**; delivered **3 technical workshops** and **2 regional events** reaching **1000+ participants**.

Event Coordination

- Coordinated **amBITion 2.0** and **AI Study Jams** for **200+ participants**; mentored **10+ junior students** on **DSA** and Python.